

miniLISA: Design and Component Noise Model of an Optical Time-Delay Interferometry Testbed

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We describe miniLISA, a laboratory testbed combining optical laser signals, electronic propagation delays and independent spacecraft clocks. The design reproduces the laser-noise structure of LISA's one-way measurements, a reduced sideband clock-transfer model, and injected gravitational-wave signals. We calculate the modulation and readout contributions introduced by the optical extension, alongside the electronic baseline and an assumed board-timing spectrum. These noises are propagated through second-generation time-delay interferometry with changing arm lengths and clock correction. The resulting sum describes the included components; it excludes residuals from the current delay implementation and does not predict the complete testbed performance. The setup will allow tests of laser- and clock-noise suppression together with recovery of an injected signal.

I. INTRODUCTION

Gravitational waves provide a unique observational window into astrophysical and cosmological phenomena. The first direct detection in 2015 established gravitational-wave observations of compact binary mergers with ground-based interferometers [1]. Accessing gravitational waves at millihertz frequencies requires a space-based detector to overcome terrestrial seismic and Newtonian noise and to provide sufficiently long interferometric baselines. The Laser Interferometer Space Antenna (LISA), a space-based observatory led by ESA with NASA participation, is designed to operate in the 0.1 mHz to 1 Hz band [2].

The LISA constellation consists of three spacecraft arranged in an approximately equilateral triangle with arm lengths of about 2.5 million km. Laser beams exchanged between the spacecraft form a constellation-scale interferometer. Unlike ground-based detectors with effectively equal arms, the LISA arm lengths vary by up to about 1.2% due to orbital motion, preventing direct cancellation of laser frequency noise. Although cavity pre-stabilisation reduces laser frequency noise substantially, it remains several orders of magnitude above the displacement noise budget required for gravitational-wave detection [3]. To suppress this noise, time delay interferometry (TDI) has been developed as a post-processing technique that synthesizes virtual equal-arm interferometers by appropriately time-shifting and combining phasemeter measurements.

In addition to laser frequency noise, LISA measurements are affected by clock noise originating from the independent ultra-stable oscillators (USOs) carried by each spacecraft. Since all phase measurements are referenced to these clocks, timing jitter couples directly into the measured heterodyne beatnotes. Space-qualified USOs achieve Allan deviations of order 10^{-13} at one-second averaging times [4], which is insufficient for the picometer-level test-mass position sensitivity required by LISA. To mitigate this effect, LISA employs additional clock-noise transfer measurements—implemented via modula-

tion sidebands exchanged between spacecraft.

TDI has been developed analytically and tested through numerical simulations [3, 5]. Simulations predict the response of a specified measurement model and test its numerical implementation. Physical experiments provide a complementary test of whether the hardware follows that model, including couplings and processing effects that an idealized simulation may omit.

The University of Florida Laser Interferometry Simulator (UFLIS) demonstrated TDI and TDI-ranging with time-varying electronic propagation delays and optical laser signals [6]. miniLISA continues this approach with FPGA delay lines and independent spacecraft clock references. The experiment of de Vine et al. demonstrated laser-noise suppression of approximately 10^9 and clock-noise suppression of 6×10^4 , using short optical paths [7]. LISA On Table demonstrated first-generation TDI with simulated noise and static unequal arms [8].

Here, we present an overview of an ongoing effort to develop a TDI testbed that combines realistic laser frequency noise, programmable delay lines, and independent clock sources within a single experimental framework. The testbed focuses on reproducing the dominant noise couplings of the LISA metrology system—specifically, laser frequency noise and clock noise—with component-noise propagation through time-dependent delays. Crucially, the same delay lines that impose the inter-spacecraft light-travel times also allow physically consistent gravitational-wave strain signals to be injected directly into the measurement chain. The testbed can therefore be used not only to demonstrate noise suppression, but to verify that a known gravitational-wave signal survives TDI processing of hardware-generated data with the expected amplitude and phase.

An initial purely electrical implementation centered around field programmable gate array (FPGA)-based digital delay lines has already been demonstrated in Ferguson et al. [9]. The developments discussed here extend that architecture by incorporating an optical front-end with physically representative laser noise and independently running spacecraft clocks. This reproduces

the laser-noise and propagation structure of the one-way measurements together with a reduced clock-transfer model, allowing TDI combinations and clock-noise calibration to be formed entirely from hardware-generated data. The goal of this work is to realize a laboratory-scale analog of the LISA inter-satellite metrology system—a “miniLISA”—that bridges the gap between idealized software simulations and the flight instrument, and provides a controlled environment for testing the robustness of TDI to instrumental non-idealities and realistic gravitational-wave signal injection.

More concretely, we look for residual laser and clock noise below the secondary-noise floor after processing, together with recovery of an injected GW signal at the expected amplitude and phase.

Section II introduces the LISA measurements and noise-suppression combinations. Section III describes their laboratory implementation, and Section IV calculates the contributions from measured and modelled noise sources. We then discuss possible extensions and remaining measurements.

II. LISA INTERFEROMETRIC DETECTION SYSTEM

We now describe the LISA interferometric detection system (IDS) and present a mathematical model of the noises. We specifically focus on relevant elements for laser frequency noise and USO jitter propagation and coupling.

A. Interferometric Phase Measurements

Each of the three identical LISA satellites carries two lasers on board. We refer to one as the local laser and the other as the adjacent laser. These two laser fields interfere on the test-mass and reference interferometers on the two optical benches of the spacecraft. In addition, the local laser interferes with the laser received from a distant spacecraft on the science interferometer. It is this science interferometer that encodes the gravitational-wave signal by measuring fluctuations in the spacecraft-to-spacecraft distance, while the test mass interferometer will measure the spacecraft-to-test mass displacement. In the upper panel of Fig. 1, a simplified LISA two-way armlink involving only the science interferometer is shown.

We begin by writing the optical phase of the laser field originating from movable optical system assembly (MOSA) ij on spacecraft i facing spacecraft j as a function of time t :

$$\Phi_{ij}(t) = \nu_{ij}t + \delta\phi_{ij}(t), \quad (1)$$

where ν_{ij} denotes the nominal carrier frequency and $\delta\phi_{ij}(t)$ represents the intrinsic laser phase noise. Throughout this section, phases are expressed in cycles, frequencies in Hz, and clock errors in seconds. Capital Φ

denotes total phase, while $\delta\phi$ denotes phase fluctuations about the nominal linear evolution. We use a single time coordinate, neglect relativistic time transformations, and retain constant nominal frequencies in the noise-coupling coefficients. The clock on spacecraft i reads $t_i = t + q_i(t)$, where q_i is its timing error.

Each of the laser fields is phase-modulated using an electro-optic modulator (EOM) driven by the local USO. This is used to imprint sidebands on the laser field that carry the USO timing noise $q_i(t)$, creating a phase noise term $\nu_{ij}^m q_i(t)$. Since there is only one USO per spacecraft, clock noise carries a single spacecraft index.

The modulation frequency ν_{ij}^m will be about 2.4 GHz, which is upconverted from the USO via a phase locked loop (PLL) synthesizer, preserving its noise with high fidelity. For the adjacent and opposing MOSAs ik and ji respectively, the corresponding modulation frequency differs by 1 MHz, so 2.401 GHz, such that the sideband-sideband beat notes are offset by 1 MHz from the carrier-carrier beat note. The modulation depth $\mu_{\text{EOM}} = 0.53 \text{ rad}$ places about 87% of the optical power in the carrier and 6.5% in each first-order sideband. The USO references the modulation, sampling-clock and pilot-tone synthesis chains; their role in clock-noise suppression is discussed in Section II C. Additional phase modulation with pseudo-random codes supports ranging and data transfer; it is distinct from the sinusoidal clock-transfer modulation considered here [10].

Propagation from MOSA ij to ji delays the emitted phase and adds the GW response along the arm. Following the separation of background propagation and GW perturbations in [3], we write

$$d_{ji}^{\text{tot}}(t) = d_{ji}(t) + H_{ji}(t). \quad (2)$$

Here $d_{ji}(t)$ is the background propagation time, which can vary slowly with the constellation breathing, and $H_{ji}(t)$ is its GW-induced perturbation. The first index denotes the receiving spacecraft and the second the emitting spacecraft. We define the delay operator using only the background propagation time,

$$\mathbf{D}_{ji}x(t) = x(t - d_{ji}(t)). \quad (3)$$

Applying the total propagation time to Eq. (1) and expanding to first order gives

$$\begin{aligned} \Phi_{ij}(t - d_{ji}(t) - H_{ji}(t)) &\simeq \nu_{ij} [t - d_{ji}(t)] \\ &\quad + \mathbf{D}_{ji}\delta\phi_{ij}(t) - \nu_{ij}H_{ji}(t), \end{aligned}$$

where the product $H_{ji}\mathbf{D}_{ji}\delta\phi_{ij}$ of the GW perturbation and laser-frequency noise has been neglected. We therefore introduce the GW phase contribution

$$\delta\phi_{ji}^{\text{GW}}(t) = -\nu_{ij}H_{ji}(t). \quad (4)$$

To this order, the received total phase is $\mathbf{D}_{ji}\Phi_{ij}(t) + \delta\phi_{ji}^{\text{GW}}(t)$: the GW signal is an additive phase contribution, separate from the background delay operator. For

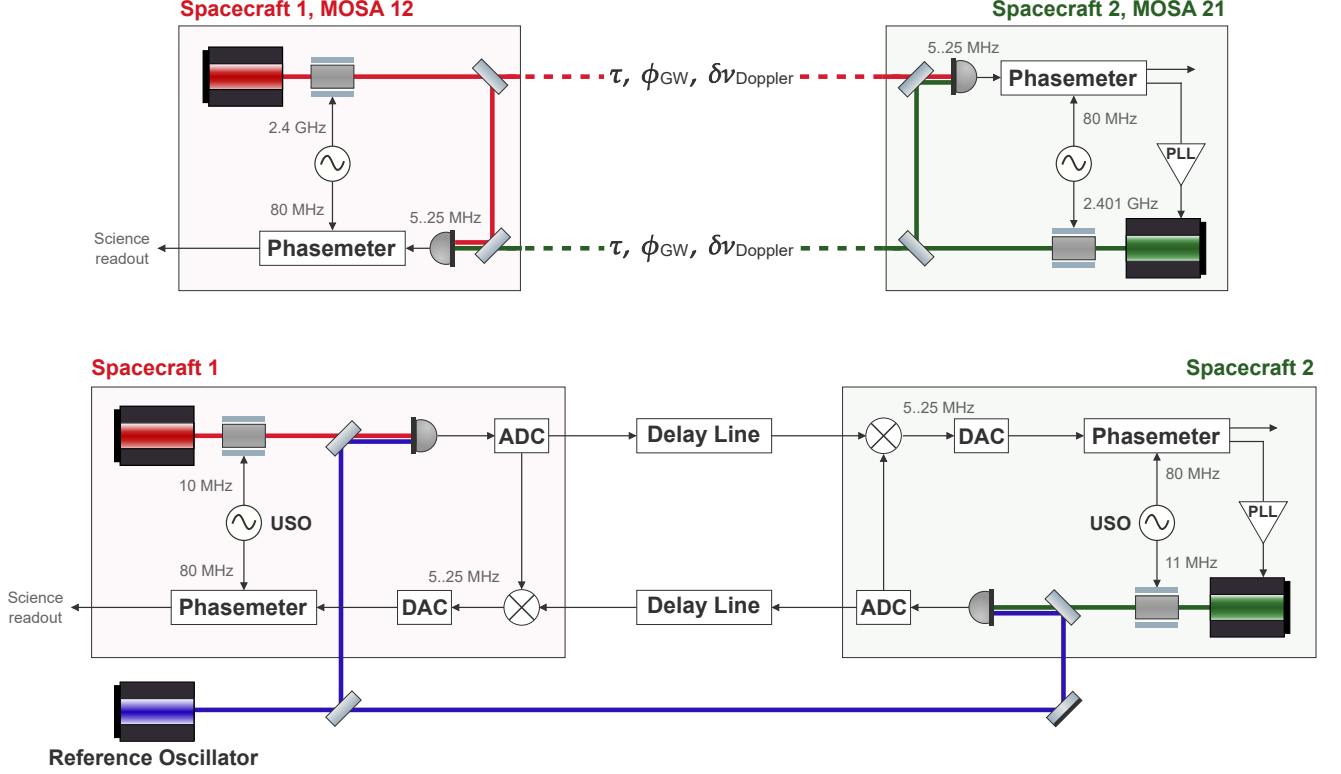


FIG. 1. A comparison of the two-way links as implemented in LISA (upper) and the testbed (lower). The upper panel shows the conceptual LISA configuration: the laser on Spacecraft 1 (MOSA 12) is phase-modulated at approximately 2.4 GHz using the local USO and transmitted to Spacecraft 2 (MOSA 21). The received beam interferes with the local laser, producing heterodyne beat notes in the MHz range that contain gravitational-wave phase fluctuations, Doppler shifts due to orbital motion, and laser frequency noise. These beat notes are tracked by the phasemeter, which is clocked by the onboard 80 MHz sampling reference derived from the USO. The lower panel depicts the laboratory implementation of the same signal topology. Instead of physical light propagation over millions of kilometres, the inter-spacecraft delay is emulated electronically using digital delay lines between ADC and DAC stages. A common reference laser is used to downconvert the laser fields from hundreds of terahertz to tens of megahertz, which can then be sampled by the delay line ADCs. The actual LISA-like heterodyne beat notes are created in the mixers.

Draft note: Revise the lower-panel labels to independently referenced Mokus, (35,36,34) MHz modulation and the synthesized output tones for all six links. The legacy USO and 10/11 MHz labels do not describe the operating point used here.

constant ν_{ij} , the corresponding frequency fluctuation is $\delta\dot{\phi}_{ji}^{\text{GW}} = -\nu_{ij}\dot{H}_{ji}$.

At spacecraft j , the received field i interferes with the local laser field of phase $\Phi_{ji}(t)$ on a photodetector. The resulting heterodyne signal contains both the carrier-carrier beat note and the sideband-sideband beat notes. Additional cross terms lie outside the heterodyne range of the phasemeter; a representative LISA frequency plan uses absolute beat frequencies from 5 to 25 MHz [11].

The electrical signal from the photodetector is then digitized by the phase measurement system (PMS) front-end electronics. The sampling process introduces a coupling of the local USO timing error $q_j(t)$ into the measured phase. For all measurement readouts, we adopt the LISA convention that phases are written as “distant

minus local.”¹ We define the nominal carrier coefficient

$$\alpha_{ji} = \Delta\nu_{ji} = \nu_{ij} - \nu_{ji}. \quad (5)$$

After removing the nominal beat evolution, the sampled carrier phase fluctuations are

$$\begin{aligned} \delta\phi_{ji}^c(t) = & \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{ji}(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) - \alpha_{ji}q_j(t). \end{aligned} \quad (6)$$

Here α_{ji} is the beat frequency in the fixed-frequency approximation used for the noise model. For an orbit-dependent frequency plan, it would be replaced by the actual signed beat frequency, including Doppler shifts.

¹ The phasemeter measures the absolute beat frequency; its sign is assigned using the frequency plan.

The signal is then passed to digital phase lock loops (DPLLs), which track the carrier and both sideband beat notes. In the following readouts, the nominal phase evolution is largely removed from the tracked phase, leaving the fluctuations used for GW detection and clock-noise suppression. To connect this noise model to the measurements used for TDI, we also include the secondary disturbances in the science readout:

$$\begin{aligned} \text{sci}_{ji}^c(t) = & \mathbf{D}_{ji} \delta\phi_{ij}(t) - \delta\phi_{ji}(t) - \alpha_{ji} q_j(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) + N_{ji}^c(t) \\ & - \frac{\nu_0}{c} (\mathbf{n}_{ij} \cdot \mathbf{D}_{ji} \mathbf{\Delta}_{ij}(t) + \mathbf{n}_{ji} \cdot \mathbf{\Delta}_{ji}(t)), \end{aligned} \quad (7)$$

The term $N_{ji}^c(t)$ represents combined readout noise from optical path-length fluctuations and electronics. The vectors $\mathbf{\Delta}_{ij}(t)$ are optical-bench displacements in metres. The unit vector $\mathbf{n}_{ji}$ points along the incoming beam from spacecraft i to spacecraft j , and $\mathbf{n}_{ij} = -\mathbf{n}_{ji}$ in the geometry used for these displacement terms. The factor $\nu_0/c = 1/\lambda_0$ converts path-length fluctuations to phase in cycles, with $\nu_0 \simeq 281$ THz for the nominal wavelength $\lambda_0 = 1064$ nm. MHz-scale offsets between lasers are small compared with this optical frequency, so a common ν_0 suffices for the displacement terms. Those offsets remain essential in the heterodyne clock coefficients, which depend on frequency differences.

The corresponding upper sideband-sideband beat note readout can be written as

$$\begin{aligned} \text{sci}_{ji}^{\text{sb},+}(t) = & \mathbf{D}_{ji} \delta\phi_{ij}(t) - \delta\phi_{ji}(t) - \gamma_{ji} q_j(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) + N_{ji}^{\text{sb},+}(t) \\ & - \frac{\nu_0}{c} (\mathbf{n}_{ij} \cdot \mathbf{D}_{ji} \mathbf{\Delta}_{ij}(t) + \mathbf{n}_{ji} \cdot \mathbf{\Delta}_{ji}(t)) \\ & + \nu_{ij}^m \mathbf{D}_{ji} q_i(t) - \nu_{ji}^m q_j(t). \end{aligned} \quad (8)$$

The sideband clock coupling coefficient is

$$\gamma_{ji} = \Delta\nu_{ji} + \nu_{ij}^m - \nu_{ji}^m = \Delta\nu_{ji} + \Delta\nu_{ji}^m = \Delta\nu_{ji}^{\text{sb},+}. \quad (9)$$

Again, the term $N_{ji}^{\text{sb},+}(t)$ represents stochastic readout noise specific to the upper sideband measurement. The lower sideband has the same laser phase noise but the opposite sign for each modulation contribution. The sidebands have optical frequencies $\nu_{ij} \pm \nu_{ij}^m$, so the same path-length change produces a slightly different phase shift in each sideband. Using ν_0/c for their displacement couplings neglects a fractional correction of order $\nu^m/\nu_0 \simeq 9 \times 10^{-6}$. Similarly, the upper-sideband GW phase is $-(\nu_{ij} + \nu_{ij}^m)H_{ji}$, differing from the carrier by $-\nu_{ij}^m H_{ji}$. We neglect this small difference in the readout equations above and use the same $\delta\phi_{ji}^{\text{GW}}$ for the carrier and both sidebands. The δ denotes a perturbation to the nominal optical phase, rather than stochastic noise.

For completeness, we also write out the reference and

test mass interferometer readings on MOSA ji

$$\begin{aligned} \text{tm}_{ji}(t) = & \delta\phi_{jk}(t) - \delta\phi_{ji}(t) + \beta_{ji} q_j(t) + \mu_j(t) + N_{ji}^{\text{tm}}(t) \\ & + \frac{2\nu_0}{c} \mathbf{n}_{ji} \cdot (\mathbf{\delta}_{ji}(t) - \mathbf{\Delta}_{ji}(t)), \end{aligned} \quad (10)$$

$$\begin{aligned} \text{ref}_{ji}(t) = & \delta\phi_{jk}(t) - \delta\phi_{ji}(t) + \beta_{ji} q_j(t) \\ & + \mu_j(t) + N_{ji}^{\text{ref}}(t). \end{aligned} \quad (11)$$

Here we have introduced the fiber noise μ_j coming from the optical fiber that exchanges the local and adjacent laser fields between MOSA ji and MOSA jk . This term is typically referred to as the backlink noise. We have assumed it to be independent of the direction of propagation of the optical beams within. The vector $\mathbf{\delta}_{ji}$ is the test-mass displacement in metres, including displacement produced by acceleration noise; it is not itself an acceleration. With the same sampling convention, the reference and test-mass clock coefficient is

$$\beta_{ji} = \nu_{ji} - \nu_{jk}. \quad (12)$$

It has the opposite sign to the adjacent-minus-local beat frequency, which explains the plus sign multiplying $\beta_{ji} q_j$ in the readouts.

These equations retain the terms needed to explain laser- and clock-noise suppression. A full instrument model includes further effects, such as tilt-to-length coupling and additional stray-light and optical-path disturbances. The miniLISA model omits the test-mass and reference interferometers and their associated bench-motion, backlink, and test-mass noise terms.

B. Laser Noise and TDI

In the LISA measurement system, laser frequency noise is first reduced onboard by locking one laser to an optical resonator and phase-locking the remaining lasers to this pre-stabilized reference with fixed frequency offsets. The last step ensures that all heterodyne beat notes remain within the phasemeter bandwidth while maintaining coherent phase relationships between spacecraft. However, residual laser frequency noise remains many orders of magnitude above the displacement sensitivity requirement and must be suppressed in post-processing. TDI achieves this by forming linear combinations of one-way phase measurements and appropriately delayed copies thereof, effectively synthesizing an equal-arm interferometer in post-processing.

For scale, consider a representative pre-stabilized frequency-noise amplitude of $30 \text{ Hz}/\sqrt{\text{Hz}}$ [3]. At Fourier frequency $f = 10 \text{ mHz}$, the corresponding single-laser phase-noise amplitude is $30/(2\pi f) \simeq 480 \text{ cycles}/\sqrt{\text{Hz}}$, equivalent to about $0.5 \text{ mm}/\sqrt{\text{Hz}}$ displacement noise at 1064 nm . This illustrative conversion is many orders of magnitude above picometre-scale displacement noise required for GW detection.

The construction proceeds in two steps: local measurements first remove optical-bench motion and reduce the six laser noises to three; delayed combinations of the resulting links then suppress the remaining laser noise [12, 13]. Using Eqs. (7), (10), and (11), we form

$$\begin{aligned} T_{ji} &= \text{tm}_{ji} - \text{ref}_{ji}, \\ \xi_{ji} &= \text{sci}_{ji}^c - \frac{1}{2} (T_{ji} + \mathbf{D}_{ji} T_{ij}). \end{aligned}$$

Within the common-optical-frequency approximation above, the bench terms cancel, leaving the test-mass contribution

$$M_{ji} = -\frac{\nu_0}{c} [\mathbf{n}_{ji} \cdot \boldsymbol{\delta}_{ji}(t) + \mathbf{D}_{ji}(\mathbf{n}_{ij} \cdot \boldsymbol{\delta}_{ij}(t))].$$

The following equations omit readout noises. Reciprocal backlink noise cancels in the local reference difference

$$z_{ji} = \frac{1}{2} (\text{ref}_{ji} - \text{ref}_{jk}) = \delta\phi_{jk} - \delta\phi_{ji} + \beta_{ji} q_j.$$

For cyclic triples $(i, j, k) = (1, 2, 3), (2, 3, 1), (3, 1, 2)$, the intermediate variables can then be defined as

$$\begin{aligned} \eta_{ji} &= \xi_{ji} - z_{ji} \\ &= \mathbf{D}_{ji} \delta\phi_{ij}(t) - \delta\phi_{jk}(t) - (\alpha_{ji} + \beta_{ji}) q_j(t) \\ &\quad + \delta\phi_{ji}^{\text{GW}}(t) + M_{ji}(t), \\ \eta_{ij} &= \xi_{ij} + \mathbf{D}_{ij} z_{ji} \\ &= \mathbf{D}_{ij} \delta\phi_{jk}(t) - \delta\phi_{ij}(t) - \alpha_{ij} q_i(t) \\ &\quad + \mathbf{D}_{ij} [\beta_{ji} q_j(t)] \\ &\quad + \delta\phi_{ij}^{\text{GW}}(t) + M_{ij}(t). \end{aligned}$$

Here M_{ij} is obtained by exchanging i and j in M_{ji} . These combinations reduce the number of independent laser phase noises from six to three, retaining $\delta\phi_{12}$, $\delta\phi_{23}$ and $\delta\phi_{31}$, while removing the optical-bench displacement terms. The construction introduces an asymmetry between the two link orientations.

For the two-arm geometry used below, we use the Michelson observable X_i . Define the synthesized round-trip measurements

$$R_{ij} = \eta_{ij} + \mathcal{D}_{ij} \eta_{ji}, \quad R_{ik} = \eta_{ik} + \mathcal{D}_{ik} \eta_{ki}.$$

The first-generation combination is

$$X_i = (1 - \mathcal{D}_{ij} \mathcal{D}_{ji}) R_{ik} - (1 - \mathcal{D}_{ik} \mathcal{D}_{ki}) R_{ij}. \quad (13)$$

Here $\mathbf{D}$ describes propagation in the measurement chain, while $\mathcal{D}$ denotes a delay applied in post-processing. Cancellation requires the processing delays to match the physical delays. A mismatch δd leaves a phase residual proportional to $\delta d \delta\phi$, so either an inaccurate delay estimate or an untracked change in the physical delay can limit laser-noise suppression.

First-generation TDI combinations cancel laser noise for constant light-travel times. Orbital motion makes

these delays time-dependent, so their order of application matters and the cancellation is no longer exact. Second-generation combinations use longer sequences of delays to cancel the leading residual from arm-length changes, leaving terms involving higher powers of the arm velocities and arm accelerations [13].

C. Clock-noise Coupling and Correction

Clock timing jitter is another primary noise in the raw measurements [2, 14]. Similarly to laser frequency noise, it exceeds the sensitivity requirement by several orders of magnitude and must be removed in post-processing before gravitational wave signals can be recovered. Unlike laser frequency noise, it couples into the measurement only locally at one spacecraft. Hence, to still allow for a similar differential measurement of the clock noise, the laser fields exchanged between the satellites are modulated with a signal derived from the USOs. From the upper-sideband and carrier phase measurements at the science interferometer, a correcting term can be constructed.

Using the phase-domain counterpart of the clock-transfer construction in [14], and neglecting readout and modulation noises initially, we define

$$r_{ji} = \frac{\text{sci}_{ji}^{\text{sb},+}(t) - \text{sci}_{ji}^c(t)}{\nu_{ij}^m} = \mathbf{D}_{ji} q_i(t) - q_j(t). \quad (14)$$

The retained GW phase cancels in this difference because we have approximated the carrier and sideband GW couplings as equal. Keeping their optical-frequency difference would instead add $-H_{ji}$ to r_{ji} ; the cancellation is therefore approximate. The normalization uses the transmitted modulation frequency ν_{ij}^m , giving r_{ji} units of seconds. Equivalently, $r_{ji} = (\text{sci}_{ji}^{\text{sb},+} - \text{sci}_{ji}^{\text{sb},-}) / (2\nu_{ij}^m)$ under the same approximations. These variables contain delayed remote clock noise minus local clock noise, matching the structure of laser noise in the science readouts. Delayed combinations of the r variables can therefore be used to construct clock corrections for the TDI observables [14]. The modulation tone must carry the same timing fluctuations that enter the phase measurement. In LISA, pilot-tone correction removes the ADC sampling jitter, making the pilot tone the relevant clock reference. Noise added between this reference and the modulation tone limits the subsequent clock correction. Let $\tau_{ij}^{\text{mod}}(t)$ denote differential timing noise in the tone derivation for MOSA ij , in seconds, and let $\psi_{ij}^{\text{mod}}(t)$ denote additional optical modulation phase noise, in cycles. The excess upper-sideband phase is

$$m_{ij}(t) = \nu_{ij}^m \tau_{ij}^{\text{mod}}(t) + \psi_{ij}^{\text{mod}}(t).$$

Including these terms and readout noise gives

$$r_{ji} = \mathbf{D}_{ji}q_i - q_j + \frac{\mathbf{D}_{ji}m_{ij} - m_{ji}}{\nu_{ij}^m} + \frac{N_{ji}^{\text{sb},+} - N_{ji}^c}{\nu_{ij}^m}.$$

Modulation and readout noise are kept separate because their transfer functions differ. A source modulation error appears in both local and delayed measurements, while a readout error is added in a particular measurement channel. Independent sources may be uncorrelated, but repeated appearances of the same source must still be combined before calculating its output noise power.

Dividing the modulation phase by its frequency gives $m_{ij}/\nu_{ij}^m = \tau_{ij}^{\text{mod}} + \psi_{ij}^{\text{mod}}/\nu_{ij}^m$. A higher modulation frequency therefore reduces the effect of a fixed phase noise, but leaves a fixed timing noise unchanged. Improving that timing-noise floor requires a quieter tone-generation chain. When the local and transmitted frequencies differ, the local timing contribution retains the factor ν_{ji}^m/ν_{ij}^m . A measurement at one modulation frequency cannot distinguish the two contributions.

The two modulation-frequency synthesis chains can add different excess noises. Local reference-interferometer sideband measurements provide additional observables for correcting the accessible differential part [14]. This contribution is absent from the present one-modulator-per-spacecraft testbed.

III. MINILISA DESIGN

A. Motivation for the Present Work

The present work extends the electronic delay-line testbed with optical inputs and independent spacecraft clocks. It combines physical laser noise, sideband-based clock transfer, and hardware delays of several seconds, and estimates the noise contributions introduced by the optical extension alongside the electronic baseline.

The electronic delays place samples acquired several seconds apart in the same interferometric measurement. This lets us test noise cancellation with independently clocked hardware, including changes in the delay caused by the hardware clock itself.

B. Measurement Philosophy

The design provides six directed links between three spacecraft, allowing LISA TDI combinations to be formed within the measurement model described below. We use the familiar Michelson X for the performance analysis; it uses four of these links. The testbed reproduces the laser frequency noise (LFN), clock-noise, and GW couplings of the LISA measurements. The first index of a link denotes the receiving spacecraft (SC); source

quantities carry a single spacecraft index, $\delta\phi_i$, q_i , and ν_i^m . The target carrier phase fluctuations are

$$\delta\phi_{ij}^c(t) = \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t) - \alpha_{ij}q_i(t),$$

and the upper-sideband fluctuations are

$$\delta\phi_{ij}^{\text{sb},+}(t) = \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t) - (\alpha_{ij} + \nu_j^m)q_i(t) + \nu_j^m\mathbf{D}_{ij}q_j(t).$$

Here α_{ij} is the signed nominal carrier beat frequency. The sideband sampling coefficient is $\gamma_{ij} = \alpha_{ij} + \nu_j^m - \nu_i^m$; combining it with the local modulation phase gives the clock term above. Both channels use the common GW phase approximation of Section II. We call a measurement *LISA-like* when it reproduces these couplings. Modulation, board, and readout noises are added below.

An FPGA delay line supplies the propagation delay and the prescribed Doppler and GW frequency shifts. Heterodyning the optical fields with a common reference laser brings their phase information into the MHz range accessible to the electronics. The reference laser noise becomes part of the effective input laser noise, whose cancellation is discussed below.

The testbed omits the test-mass and reference interferometers, together with test-mass acceleration noise, optical-bench motion, backlink noise, and tilt-to-length coupling. Consequently, the local combinations T_{ji} and ξ_{ji} used to remove bench motion in Section II are unnecessary. Identifying the two bench lasers on each spacecraft with a single source also removes the need for the reference differences z_{ji} . The synthesized links reproduce the three-laser structure of the η observables, but omit the reference-interferometer clock terms proportional to β_{ji} . The clock correction tested here is therefore the reduced case of the LISA construction with those terms absent.

C. Optical Input

Three lasers represent the spacecraft, each with an independent clock driving its modulation and phase measurement. A fourth laser provides the common optical reference. The role of the reference laser is to convert the optical phase fluctuations of each SC laser into electronically processable heterodyne beat notes, which can then be measured by the delay line. This heterodyning step effectively downconverts the optical signals from approximately 281 THz to tens of megahertz, preserving laser frequency noise in the millihertz range of interest. Among the SC lasers, one is frequency-stabilized to a 25 cm ultra-low expansion (ULE) glass cavity, while the reference laser is locked to a 30 cm ULE cavity via the Pound-Drever-Hall method. Both cavities are housed in a vacuum chamber surrounded by three layers of aluminum thermal shielding. The target is a pre-stabilized LFN level of around 30 Hz/ $\sqrt{\text{Hz}}$ in the millihertz band.

The remaining two SC lasers are PLL-locked to the synthetic LISA-like beat notes.

Each spacecraft uses an independent timing reference for its Moku. The Moku supplies both the phase readout and the modulation tone driving the fiber-coupled EOM. We use q_i for the timing fluctuations common to the EOM drive and the phasemeter's sampling reference. Excess noise between these paths enters the modulation and readout terms separately.

The reference laser is unmodulated in the configuration described here, so the spacecraft carrier and sidebands each beat with the reference carrier. The modulation frequency must therefore keep all three electrical beat notes within the delay-line input bandwidth. These input tones carry the optical phase and clock information; the LISA-like carrier-carrier and sideband-sideband readouts are formed by digital subtraction downstream. Modulating the reference laser is a possible extension for higher modulation frequencies, discussed in Section V C. The construction of a two-way arm link is illustrated in the lower panel of Fig. 1, alongside a simplified schematic of the corresponding LISA link showing only the functionally relevant elements. Taking the signed photodetector beat phase as SC laser minus reference laser, the carrier signal phase before additive detector noise is

$$\Phi_{PD,i}^c(t) = \nu_{R,i}t + \delta\phi_i^{\text{eff}}(t),$$

where $\nu_{R,i}$ is the signed nominal beat note frequency between the reference laser and the SC i laser, and $\delta\phi_i^{\text{eff}} = \delta\phi_i - \delta\phi_R$ includes their relative laser phase noise. The upper-sideband phase is

$$\begin{aligned} \Phi_{PD,i}^{\text{sb},+}(t) = & (\nu_{R,i} + \nu_i^m)t \\ & + \delta\phi_i^{\text{eff}}(t) + \nu_i^m q_i(t) \\ & + \nu_i^m \tau_i^{\text{mod}}(t) + \psi_i^{\text{mod}}(t). \end{aligned}$$

Here, τ_i^{mod} is timing noise added by the modulation path relative to the common reference q_i , while ψ_i^{mod} is additional modulation phase noise. Noise added only by the sampling path is included in the readout terms below. These are the single-index counterparts of τ_{ij}^{mod} and ψ_{ij}^{mod} in Section II C, with the same dependence on modulation frequency. We write their combined upper-sideband phase noise as

$$m_i(t) = \nu_i^m \tau_i^{\text{mod}}(t) + \psi_i^{\text{mod}}(t).$$

The lower sideband carries the opposite modulation phase, $-\nu_i^m q_i - m_i$. If its signed electrical frequency $\nu_{R,i} - \nu_i^m$ is negative, representing it at positive frequency also reverses the measured phase sign; the clock correction below uses the upper sideband.

The photodetector output is then split and distributed across three delay-line boards. One of these boards synthesizes the local LISA-like beat note signals, the other two do so for the other two SC. The signal routing between the boards is illustrated in Fig. 2.

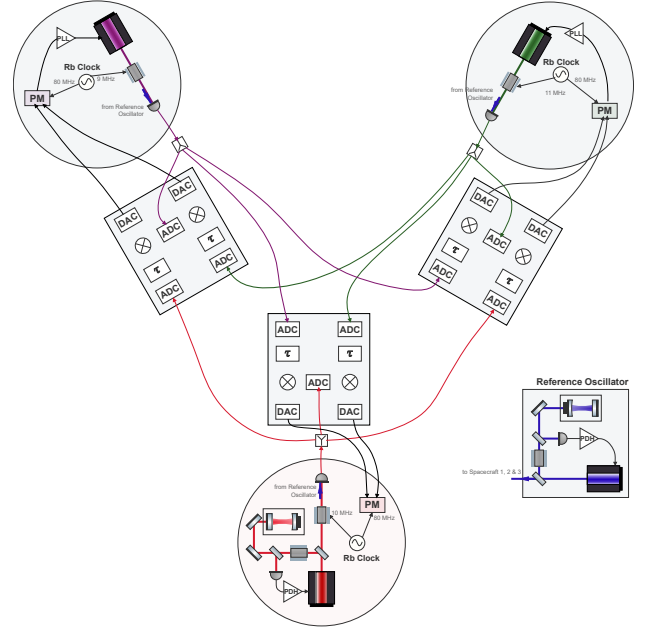


FIG. 2. Planned six-link routing architecture for miniLISA. Each spacecraft has an optical source and an independently referenced Moku phasemeter that also supplies its EOM drive; each delay-line board can synthesize two incoming links. The complete routing supports six directed links, whereas the Michelson calculation in Section IV uses links 12, 21, 13 and 31. The spacecraft-1 and reference lasers are stabilized to separate optical cavities; the other spacecraft lasers are phase-locked to synthesized beat notes. The common reference laser supplies the optical heterodyne reference.

Draft note: Update the schematic labels from the legacy modulation frequencies to (35, 36, 34) MHz and identify the independent spacecraft clock references.

Shot noise, intensity noise and detector electronics add fluctuations to the photocurrent, which produce phase errors when the delay-line PLLs track the beat notes. Noise added before the electrical split is shared between boards; ADC and electronics noise added after the split need separate terms. Their phase-noise scales are calculated in Section IV B, and their different propagation is described in Section IV D.

D. The Delay Line

The digital delay line emulates the light-travel time along each directed link. The current firmware always tracks and stores the carrier and both sidebands. At the resulting data rate, the available memory supports a maximum delay of approximately 8.5 s, covering LISA's nominal one-way light-travel time of about 8.3 s.

The implementation uses an AMD Xilinx ZCU208 RF-SoC evaluation board, with integrated ADCs and DACs, FPGA programmable logic, and an embedded processing unit (PS). For each synthesized link, separate DPLLs

track the carrier and sidebands of the remote and local input signals. The tracked remote frequencies are buffered to impose the arm delay, then combined with the current local frequencies. The present 125 MHz input-processing rate restricts the optical heterodyne tones to below 62.5 MHz. This is a constraint on the laboratory optical frequency plan, distinct from the frequencies of the synthesized LISA-like outputs.

Orbital delays and GW link responses are computed in advance in Python and uploaded to the PS. During playback, the PS converts the model inputs into fixed-point commands and streams timestamped records to the programmable logic. The buffer supplies the macroscopic light-travel time, while fine delay actuation follows the prescribed arm-length evolution using fractional-delay interpolation. The underlying electronic architecture is described in [9].

The FPGA subtracts the local frequency from the delayed remote frequency and adds the prescribed Doppler and GW offsets. Integrating this frequency produces the output phase for each of the carrier and sideband tones, which are combined at the DAC. Section III E describes the GW scaling. Omitting the nominal frequency evolution, the intended carrier phase is

$$\mathbf{D}_{ij}\delta\phi_j^{\text{eff}} - \delta\phi_i^{\text{eff}} + \delta\phi_{ij}^{\text{GW}}.$$

Section IV D propagates the component noises through ideal phase delays with changing arm lengths.

The two ADC inputs and the DAC output share the timing jitter $\epsilon_i(t)$ of the board on spacecraft i . Sampling imprints this jitter on both inputs. The remote contribution then passes through the delay, while subtracting the local signal in the digital mixer reverses the sign of its ADC contribution. The DAC adds jitter at the output beat frequency α_{ij} , giving

$$\begin{aligned}\delta\phi_{\epsilon,ij}^c(t) &= -\nu_{R,j}\mathbf{D}_{ij}\epsilon_i(t) \\ &\quad + \nu_{R,i}\epsilon_i(t) + \alpha_{ij}\epsilon_i(t) \\ &= \nu_{R,j}(1 - \mathbf{D}_{ij})\epsilon_i(t).\end{aligned}$$

Here $\alpha_{ij} = \nu_{R,j} - \nu_{R,i}$, so the local ADC and DAC terms combine to give $\nu_{R,j}\epsilon_i(t)$.

The same cancellation applies to the upper sideband, with the remote input frequency replaced by $\nu_{R,j} + \nu_j^m$:

$$\delta\phi_{\epsilon,ij}^{\text{sb},+}(t) = (\nu_{R,j} + \nu_j^m)(1 - \mathbf{D}_{ij})\epsilon_i(t).$$

Both expressions assume constant nominal frequencies and no additional output frequency offset.

The same board jitter also changes the physical duration of a programmed delay. For a constant command d_{ij} in board time $u_i(t) = t + \epsilon_i(t)$, the input and output times satisfy $u_i(t) - u_i(t_{\text{in}}) = d_{ij}$. To first order,

$$d_{ij}^{\text{phys}}(t) = t - t_{\text{in}} = d_{ij} + \delta d_{ij}^{\epsilon}(t), \quad (15)$$

$$\delta d_{ij}^{\epsilon}(t) = -(1 - \mathbf{D}_{ij})\epsilon_i(t). \quad (16)$$

Thus the physical delay can fluctuate even when its programmed value is known exactly. In LISA, ranging errors leave uncertainty in the light-travel times used for TDI. Here, board timing changes the realized delay itself. Both effects can leave laser noise if the delays applied in processing do not match the physical delays.

E. GW Signal Injection

The GW phase defined in Eq. (4) is implemented electronically after the background delay. The required phase on the link from j to i is

$$\delta\phi_{ij}^{\text{GW}}(t) = -\nu_j H_{ij}(t).$$

Since the nominal optical frequency ν_j is constant, the frequency offset supplied to the phase accumulator is

$$\delta\nu_{ij}^{\text{inj}}(t) = \delta\dot{\phi}_{ij}^{\text{GW}}(t) = -\nu_j \dot{H}_{ij}(t). \quad (17)$$

Accumulating this offset reproduces the required GW phase up to an initial phase constant. The scale factor is the represented optical frequency, not the MHz beat note frequency entering the delay-line ADC. Thus the GW signal is generated by adding phase evolution to the reconstructed signal; the hardware delay itself remains d_{ij} and is not additionally perturbed by H_{ij} . This implements the first-order propagation model of Section II.

Each directed-link waveform is evaluated at its reception time t . The propagation and source geometry are already included in H_{ij} , so no additional background delay is applied to the injection waveform after it is generated. Relative timing between the playback streams must reproduce this convention.

The same GW offset is applied to the carrier and both sidebands, following the approximation in Section II. It therefore cancels from their difference used for clock correction while remaining in the science signal.

When the link response is supplied as a fractional-frequency signal $y_{ij} = -\dot{H}_{ij}$, the command is $\delta\nu_{ij}^{\text{inj}} = \nu_j y_{ij}$, using $\nu_j \simeq c/(1064 \text{ nm})$ for the optical scale. The link responses can be computed in Python for a chosen waveform and constellation geometry. Multiple sources can be combined by summing their link responses in the linear regime. The source class is not fixed by the hardware, but the resulting commands must lie within its frequency range, resolution and update bandwidth.

A signed 32-bit GW command is added on the 64-bit phase-increment scale. At a 125 MHz accumulator clock, one least significant bit corresponds to $125 \text{ MHz}/2^{64} \simeq 6.78 \times 10^{-12} \text{ Hz}$.

Draft note: Generate the delay commands and GW responses from the same orbit and reception-time convention. State the command update rate and verify the recovered amplitude and phase, including quantization and playback timing.

F. Readout and Data Processing

The electrical signals produced by the delay line are digitized and tracked by Moku phasemeters, with each spacecraft's phase measurements referenced to its own clock.

The synthesized links can be used directly as η observables. Adding in the modulation, board, and readout noises, the carrier measurement is

$$\eta_{ij}(t) = \mathbf{D}_{ij}\delta\phi_j^{\text{eff}}(t) - \delta\phi_i^{\text{eff}}(t) + \delta\phi_{ij}^{\text{GW}}(t) - \alpha_{ij}q_i(t) + \nu_{\text{R},j}(1 - \mathbf{D}_{ij})\epsilon_i(t) + b_{ij}^c(t),$$

and the corresponding upper-sideband–upper-sideband beat note as

$$\begin{aligned} \eta_{ij}^{\text{sb},+}(t) &= \mathbf{D}_{ij}\delta\phi_j^{\text{eff}}(t) - \delta\phi_i^{\text{eff}}(t) + \delta\phi_{ij}^{\text{GW}}(t) - \gamma_{ij}q_i(t) \\ &\quad + \nu_j^m \mathbf{D}_{ij}q_j(t) - \nu_i^m q_i(t) \\ &\quad + \mathbf{D}_{ij}[\nu_j^m \tau_j^{\text{mod}}(t) + \psi_j^{\text{mod}}(t)] \\ &\quad - \nu_i^m \tau_i^{\text{mod}}(t) - \psi_i^{\text{mod}}(t) \\ &\quad + (\nu_{\text{R},j} + \nu_j^m)(1 - \mathbf{D}_{ij})\epsilon_i(t) + b_{ij}^{\text{sb},+}(t). \end{aligned}$$

The terms proportional to q_i describe timing common to the modulation and phasemeter reference, while those proportional to ϵ_i come from the delay-line board.

The additive phase noise b_{ij}^a combines optical readout, residual board, and final-phasemeter noise, in cycles; timing and modulation noise are included explicitly. To retain the noise shared by copies of a photodetector signal, write

$$b_{ij}^a = \mathbf{D}_{ij}p_j^a - p_i^a + e_{ij}^a, \quad (18)$$

where p_i^a is the equivalent input phase noise in channel a before the split, and e_{ij}^a contains noise added by the individual electronic paths. This expression assumes matched tracking responses. Any mismatch must be included in the transfer functions rather than treating copies of p_i^a as independent sources.

The effective laser phases $\delta\phi_i^{\text{eff}}$ enter with the same delayed-difference structure as the LISA laser phases. TDI therefore suppresses both the spacecraft and reference laser noise.

Finally, the upper-sideband-minus-carrier variable takes the form

$$\begin{aligned} r_{ij} &= \frac{\eta_{ij}^{\text{sb},+} - \eta_{ij}}{\nu_j^m} \\ &= \mathbf{D}_{ij}q_j - q_i + \frac{\mathbf{D}_{ij}m_j - m_i}{\nu_j^m} \\ &\quad + (1 - \mathbf{D}_{ij})\epsilon_i + \frac{b_{ij}^{\text{sb},+} - b_{ij}^c}{\nu_j^m}. \end{aligned}$$

The common GW phase cancels, leaving the clock-transfer signal together with modulation, board and readout noise.

The following common-mode cancellation assumes a shared board timing reference, with residual differences between distribution paths. Writing $\epsilon_i = \epsilon_{\text{com}} + \Delta\epsilon_i$ separates their shared fluctuation from the differential jitter. In the constant-frequency model above, the common term can be absorbed into the effective laser phases and clock errors as $\delta\phi_i^{\text{eff}} - \nu_{\text{R},i}\epsilon_{\text{com}}$ and $q_i - \epsilon_{\text{com}}$. It is therefore suppressed with laser noise by TDI and clock correction. Differential board jitter remains and sets the relevant timing residual.

Draft note: State which oscillator references the delay-board ADCs, DACs and delay counters, and how it is distributed. Confirm whether the boards share one reference or follow their spacecraft clock references. In the latter case, specify the correlations between ϵ_i and q_i and revise the timing budget accordingly.

Tables I and II summarize the noise inputs. The optical readout calculation and the measured electronic and modulation spectra are discussed in Section IV.

The science-readout allocations are IRD2-IDS-04010 and IRD2-IDS-04580. Their ASDs rise as $\sqrt{1 + (f_k/f)^4}$, with $f_k = 2\text{ mHz}$ for the carrier and 0.7 mHz for the sideband. The optical-metrology comparison curve used later includes a different, total noise allocation.

TABLE I. LISA noise scales and the corresponding miniLISA inputs. Laser and readout allocations are plateau values; clock and measured component levels are quoted at 10 mHz. Predictions and experimental targets are not measured performance. The frequency dependence and measurement assumptions are given in Section IV.

Noise source	LISA	miniLISA	Treatment
Sources included in both			
Laser frequency noise	30 Hz/ $\sqrt{\text{Hz}}$ reference	Same reference adopted	Suppressed by TDI.
Spacecraft clock timing q_i	10.1 ps/ $\sqrt{\text{Hz}}$ model	Same level targeted	Sideband clock correction.
Modulation noise	No separate level assigned here	4.0×10^{-6} cycles/ $\sqrt{\text{Hz}}$ measured estimator	To be remeasured; contributes through clock correction.
Carrier readout	9 pm/ $\sqrt{\text{Hz}}$ allocation	0.0036 pm/ $\sqrt{\text{Hz}}$ calculation	Bright-beam input readout, including ideal ADC quantization.
Sideband readout	600 $\mu\text{rad}/\sqrt{\text{Hz}}$ allocation	0.0778 $\mu\text{rad}/\sqrt{\text{Hz}}$ calculation	Shared detector inputs have different propagation from independent link readouts.
ADC / phasemeter noise	Included in readout allocations	Electronic baseline in Table II	Not added twice in the combined curve.
LISA sources not reproduced in miniLISA			
Test-mass acceleration	3 fm s $^{-2}/\sqrt{\text{Hz}}$ reference scale	No free test masses	Used only in the comparison curve.
Optical-bench motion	No separate level used here	Not reproduced	Removed by LISA's local combinations.
Backlink noise	No separate level used here	No inter-bench backlink	Reciprocal part cancels in LISA.
Test-mass / reference readout	No separate levels used here	No corresponding interferometers	Absent from the reduced model.
Tilt-to-length noise	No separate level used here	Not explicitly modelled	Laboratory alignment noise is not quantified.

TABLE II. Additional sources associated with the laboratory implementation. Levels at 10 mHz are illustrative input values. The electronic spectrum is measured; the board-timing spectrum and its independence between boards are assumptions.

Noise source	Level used	Treatment
Shared reference laser	Included in effective laser phases	Shared between all optical inputs; suppressed with laser noise by TDI.
Delay-board timing ϵ_i	2.25 ps/ $\sqrt{\text{Hz}}$ Rb-model estimate	Propagated for independent boards without additional timing correction. A common fluctuation cancels to the same order as laser noise.
Electronic baseline	4.7×10^{-6} cycles/ $\sqrt{\text{Hz}}$ measured	Assigned to the carrier and sideband link readouts.
Delay-implementation residuals	Not quantified in this calculation	Development is ongoing; excluded from the component sum.

IV. NOISE CONTRIBUTIONS

A. Laser and Clock Noise

The optical extension adds modulation and detector noise to the electronic testbed. We estimate these contributions, combine them with the measured electronic baseline and an assumed board-timing spectrum, and propagate them through TDI with changing arm lengths. The result describes the included noise sources. It is not a prediction of the complete testbed performance, since work on the delay implementation is ongoing.

For the laser frequency noise we adopt the LISA pre-stabilization reference [3],

$$A_\nu(f) = 30 \frac{\text{Hz}}{\sqrt{\text{Hz}}} \sqrt{1 + \left(\frac{2 \text{ mHz}}{f}\right)^4}. \quad (19)$$

For the spacecraft clocks we use the LISA fractional-frequency noise model [14],

$$S_{\dot{q}}(f) = 4 \times 10^{-27} \left(\frac{f}{1 \text{ Hz}} \right)^{-1} \text{ Hz}^{-1}. \quad (20)$$

The timing-noise ASD is $\sqrt{S_{\dot{q}}}/(2\pi f)$. The experiment will be configured to reproduce this clock-noise level over the analysis band. It refers to the timing shared by each spacecraft's modulation and phase measurement, not to the separate delay-board jitter ϵ_i .

These are the primary noises that TDI and clock correction are intended to suppress. The calculation below assumes that the processing delays match the intended physical delays. It does not use these input levels to predict residuals from the current hardware's delay errors.

B. Modulation and Optical Readout Noise

The modulation measurement takes half the phase difference between the upper and lower sidebands of a local optical beat. This removes the common optical phase and retains modulation fluctuations together with noise added by the readout. We use the present spectrum as an estimate of the source modulation noise m_i . The measurement will be repeated with the planned modulation drive to update this estimate and assess the readout contribution.

For the detector calculation, one phase-modulated beam interferes with an unmodulated reference. Each beam delivers 4 mW at the detector and the modulation depth is 0.53 rad. The assumed responsivity is 0.6 A/W and the heterodyne power-overlap efficiency is 0.9, giving carrier and first-order sideband peak currents of 4.24 and 1.16 mA.

Shot noise, dark-current shot noise, thermal noise and additive intensity noise are converted into phase noise by dividing by the beat amplitude. For a locally white current-noise ASD i_n and peak beat current I_{pk} ,

$$A_\phi = \frac{\sqrt{2} i_n}{2\pi I_{\text{pk}}}. \quad (21)$$

The factor $\sqrt{2}$ accounts for noise on both sides of the RF beat; 2π converts radians to cycles. With an ideal three-way electrical split and a 14-bit, 2 GS/s ADC, the calculated carrier and sideband readout ASDs are 3.40×10^{-9} and 1.24×10^{-8} cycles/ $\sqrt{\text{Hz}}$. The ADC sampling rate here is distinct from the 125 MHz digital processing rate. Figure 3 shows the contributions.

These small values follow from the strong laboratory beats and ideal readout assumptions. They are not measurements of the complete receiver. In particular, RF losses, excess ADC noise and imperfect filtering can raise the readout contribution. The calculation retains additive intensity noise around each beat; additional intensity-to-phase coupling is not characterized here [15].

miniLISA does not need to reproduce LISA's weak received light or its shot-noise-limited readout. Stronger signals give a lower detector contribution and make residual laser and clock noise easier to study. With the present inputs, the electronic and modulation spectra exceed the calculated detector noise.

Detector noise added before the electrical split is shared between boards. It must therefore be propagated as a shared input, while noise added after the split is assigned to the individual readouts. This differs from adding an independent detector noise to every LISA link.

C. Electronic Noise and Board Timing

The electronic delay line and its timing-noise coupling were studied by Ferguson et al. [9]. We retain the measured electronic baseline as an input to the optical-testbed calculation. The same baseline PSD is assigned to carrier and sideband readouts, with independent equivalent noise terms for each link and channel. This represents the measured chain without separating every electronic component. The ideal ADC estimate above is not added again to the combined curve.

Development of the delay line is ongoing. The additive baseline and the coupling of board timing remain relevant when optical inputs are introduced, so both are included here. Other residuals associated with the present delay implementation are being addressed separately. Their eventual reduction has not been demonstrated by this calculation.

For board timing only, we use the Rb-standard model: a 10 MHz reference with single-sideband phase noise of -130 dBc/Hz at 10 Hz [16]. Below 10 Hz, its phase PSD is extrapolated as f^{-1} down to 1 Hz, f^{-2} down to 10 mHz, and f^{-3} below 10 mHz. This gives a timing ASD of $2.25 \text{ ps}/\sqrt{\text{Hz}}$ at 10 mHz. It is an assumed board-clock spectrum, separate from the LISA spacecraft-clock model above.

The plotted board contribution assumes independent ϵ_i and applies no additional board-timing correction. A fluctuation shared by all boards cancels to the same order as laser noise in the ideal model; differences between boards can remain. The actual contribution therefore depends on the clock distribution. Board timing also changes the physical delay, as described in Section IIID. Its nominal beat-frequency coupling is already included in the ϵ_i terms and is not added again as independent ranging noise. The coupling of delay fluctuations to laser-frequency noise is not included in this component sum.

D. Propagation Through TDI with Changing Arms

We use the four links of the second-generation Michelson observable X_2 . Every delay is evaluated at the time reached after the preceding delays. For example,

$$\mathcal{D}_{12}\mathcal{D}_{21}x(t) = x(t - d_{12}(t) - d_{21}(t - d_{12}(t))). \quad (22)$$

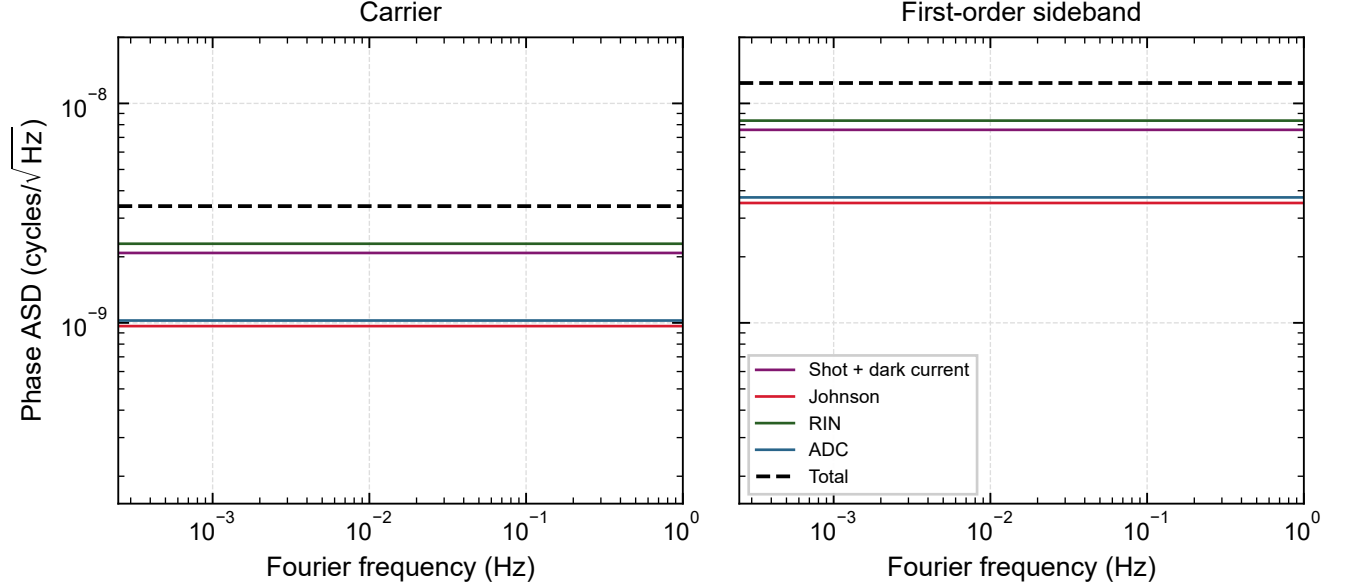


FIG. 3. Calculated carrier and first-order sideband readout noise for the stated bright-beam operating point. The model uses a $50\ \Omega$ load at 300 K, 100 nA dark current, and white fractional intensity noise of $9 \times 10^{-9}/\sqrt{\text{Hz}}$, with the two beams' additive intensity noise summed coherently. The assumed current-to-voltage gain at each ADC is $50/\sqrt{3}\ \text{V/A}$, with a 1 V full-scale span, 14 bits and 2 GS/s. The ADC curve is ideal quantization noise. The calculation assumes linear detection and adequate filtering before sampling and decimation; it does not include a measured receiver or phasemeter noise floor.

Changing the delay order generally changes the result. The calculation retains this ordering rather than multiplying the first-generation result by a static filter.

The example uses $d_{12}(t) = d_{21}(t) = L + vt$ and $d_{13}(t) = d_{31}(t) = L - vt$, with $L = 8.339\ \text{s}$ and $v = 5 \times 10^{-8}\ \text{s/s}$. One arm grows while the other shrinks. This is a simple flexing-arm example, not an orbit reconstruction. The nominal electrical input frequencies are (15.0, 10.9, 21.6) MHz and the modulation frequencies are (35, 36, 34) MHz. The beat-frequency coefficients are held constant to isolate the effect of changing delays.

For this reduced measurement model, $\alpha_{j1} = -\alpha_{1j}$. The clock terms cancel in the corrected round trips

$$R_{1j}^c = \eta_{1j} + \mathcal{D}_{1j}\eta_{j1} - \alpha_{1j}r_{1j}, \quad j = 2, 3. \quad (23)$$

The corrected second-generation combination is then

$$X_{2c} = (1 - \mathcal{D}_{121} - \mathcal{D}_{12131} + \mathcal{D}_{1312121})R_{13}^c - (1 - \mathcal{D}_{131} - \mathcal{D}_{13121} + \mathcal{D}_{1213131})R_{12}^c. \quad (24)$$

Successive subscripts specify the ordered link path, for example $\mathcal{D}_{12131} = \mathcal{D}_{12}\mathcal{D}_{21}\mathcal{D}_{13}\mathcal{D}_{31}$. This is the phase-domain Michelson construction of [14]. It cancels laser noise through first order in the arm-length rates. The simple round-trip clock correction used here relies on constant, opposite forward and reverse beat frequencies; an orbit-dependent frequency plan requires the corresponding general clock correction.

Each noise source is inserted into the carrier and sideband measurements before forming Eq. (24). All appearances of one source are combined before calculating its

noise power. This retains the correlations of shared detector inputs and of the carrier readout reused in clock correction. Independent source powers are then added. Four independent noises with unit weights would give twice the single-source ASD; the actual TDI weights depend on frequency.

For the spectral curves, we evaluate the response of each ordered path at 25 times spanning one day and average the local output powers. This uses the slow variation of the delays to estimate the component spectra; it is not a full time-domain simulation of an orbit or of the firmware. The calculation is checked against the static limit and the first-order laser-noise cancellation for unequal arm rates.

E. Combined Noise Contributions

Figure 5 combines the modulation estimate, calculated detector noise, measured electronic baseline and assumed board jitter. Their propagated powers are added to obtain the black curve. It shows the noise associated with these inputs under the stated model, rather than the performance expected from the current delay-line implementation.

The dashed reference uses the usual optical-metrology displacement scale of $15\ \text{pm}/\sqrt{\text{Hz}}$ and test-mass acceleration scale of $3\ \text{fm s}^{-2}/\sqrt{\text{Hz}}$, with their frequency dependence as specified in [2]. Both are converted to phase at

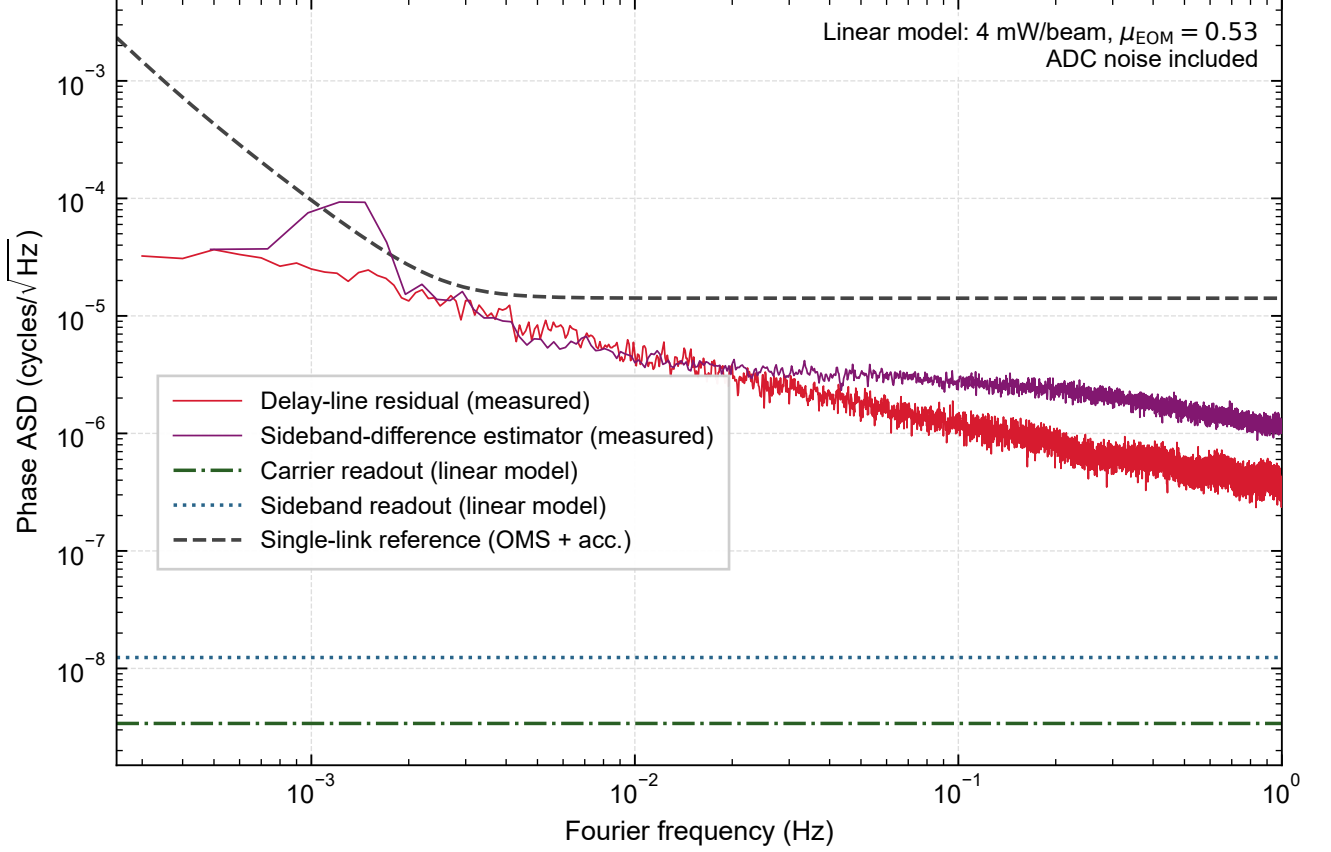


FIG. 4. Input component spectra before TDI and clock correction. The measured electronic residual and sideband-difference estimator are plotted without smoothing over 0.25 mHz–1 Hz. The calculated carrier and sideband readout use the operating point of Fig. 3, including ideal ADC quantization. The dashed curve is the single-link OMS-plus-acceleration comparison. The separate board-timing contribution is included after propagation in Fig. 5.

1064 nm and propagated through the same science combination. This is a comparison of noise scales; test-mass noise is not physically present in miniLISA.

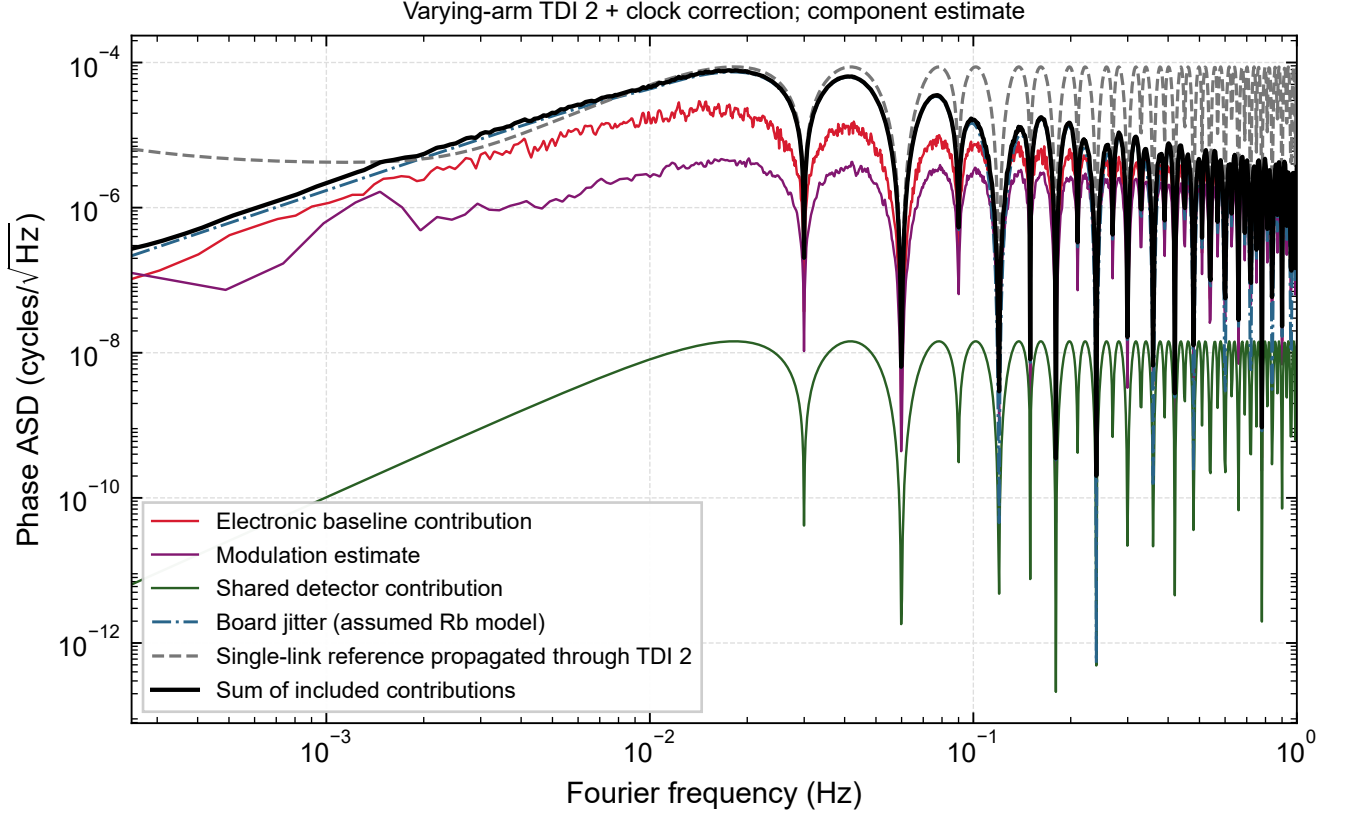
The optical readout remains a small contribution for the assumed bright-beam operating point. In this example, independent board timing gives a larger contribution than the electronic baseline over much of the low-frequency band. That conclusion depends on the assumed independence and the absence of additional board-timing correction. The repeat modulation measurement will update the optical-extension estimate. Residuals from the delay implementation must be considered separately before comparing this component sum with the complete experiment.

V. FUTURE EXTENSIONS

A. Injection of Secondary Noise

Test-mass displacement can be emulated by adding the corresponding phase terms M_{ij} from Section II to the synthesized links. Each simulated test mass must use the same displacement realization wherever it occurs, including the delayed occurrence in the reverse link. Independent noise added to every link would not reproduce these correlations. For a specified acceleration spectrum, the displacement ASD is obtained by dividing by $(2\pi f)^2$ over the analysis band. This injection would test the secondary-noise transfer functions as well as primary-noise suppression.

Independent phase perturbations can also be added after link synthesis to test a prescribed LISA-like readout-noise model. Lower optical power would test weak-light tracking and readout, but would not by itself remove the correlations introduced by splitting one detector output across boards.



Modulation (SC1,2,3): 35, 36, 34 MHz; electrical carriers: 15, 10.9, 21.6 MHz
 Linear model: 4 mW/beam, $\mu_{\text{EOM}} = 0.53$; ADC covered by electronic baseline

FIG. 5. Included noise contributions after ordered, varying-arm TDI 2 and clock correction. The black curve is their quadrature sum. The board-jitter curve uses independent Rb-model timing fluctuations; it is not measured board performance. The calculation uses the changing delays and constant frequency plan of Section IV D. The dashed reference is the OMS-plus-acceleration model propagated as four independent equivalent link noises. Delay-implementation residuals are excluded, so the sum is not a sensitivity prediction for the assembled testbed.

B. Orbit-Based Delays

The calculation in Section IV D uses a simple model with one growing arm and one shrinking arm. Orbit-derived delays and Doppler shifts would extend it to a full constellation model. The six-link design also supports combinations beyond Michelson X_2 .

For the frequency-tracking delay line to reproduce a delayed phase, its output must include the delay-rate factor,

$$\frac{d}{dt}\phi(t - d(t)) = (1 - \dot{d}(t))\dot{\phi}(t - d(t)). \quad (25)$$

Implementing and testing this response is part of the ongoing delay-line work. The noise calculation assumes the intended phase response. A full orbit model must also account for changing beat frequencies in the clock correction and keep the synthesized tones within the readout bandwidth.

C. Changes to the Clock-Transfer and Locking Configuration

Changing the modulation frequency requires separating the measured estimator into modulation, timing and readout contributions, as discussed in Section II C. A GHz implementation would also require a compatible optical and RF frequency plan. Modulating the reference laser would allow corresponding optical sidebands to beat within the electronic input band.

A second optical source and modulator on each spacecraft, together with reference interferometers and backlinks, would reproduce the local $\xi/z/\eta$ reduction and the β clock couplings omitted here [12]. Distinct synthesis chains and a local sideband comparison would also allow tests of differential tone-generation noise.

VI. CONCLUSION

miniLISA is being developed to test whether laser and clock noise can be removed from laboratory measurements while preserving a gravitational-wave signal. The design combines optical signals, independent spacecraft clocks and electronic delays that represent the light-travel times between spacecraft. Its six directed links will support different TDI combinations. Here, we use the four links needed for the second-generation Michelson combination to calculate the included noise contributions.

The calculation combines component measurements with optical readout and board-timing noise models. For the assumed operating conditions, the detector contribu-

tion is smaller than the electronic and modulation contributions. miniLISA does not need to reproduce LISA's weak-light readout: strong laboratory signals make it easier to study the remaining noise after laser and clock correction. The component noises are propagated with changing delays. Residuals from the current delay implementation are excluded, so the combined curve does not predict the performance of the complete setup.

The next steps are to repeat the modulation-noise measurement and assemble the optical setup. The main experimental test will then be to suppress laser and clock noise and recover an injected gravitational-wave signal with the expected amplitude and phase.

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